

SAINT-GERMAIN-DE-GRANTHAM, QC, Canada

WMO: 719670

Lat: 45.8253N

Lon: 72.5369W

Elev: 86

StdP: 100.30

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.6	(c) -23.6	(d) -30.0	(e) 0.2	(f) -26.1	(g) -27.1	(h) 0.3	(i) -23.2	(j) 13.7	(k) -4.1	(l) 12.0	(m) -6.9	(n) 1.9	(o) 140	(p) 0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	29.2	22.0	27.8	20.9	26.4	20.3	23.4	27.4	22.4	26.2	21.4	25.0	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.0	25.8	21.1	15.9	24.5	20.1	15.0	23.5	70.1	27.1	66.3	26.1	62.7	24.9	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.1	8.1	-32.6	31.7	3.3	0.8	-34.9	32.3	-36.9	32.8	-38.7	33.3	-41.1	33.9
			-32.7	25.0	3.3	1.1	-35.0	25.8	-36.9	26.4	-38.8	27.0	-41.1	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-10.0	-8.5	-3.1	5.4	13.1	17.2	20.1	18.9	15.0	8.3	1.3	-5.7		
	DBStd	11.79	7.50	6.78	6.66	4.56	4.33	3.52	2.96	2.83	4.31	4.59	5.31	7.14		
	HDD10.0	2562	621	519	407	152	17	1	0	0	6	89	262	488		
	HDD18.3	4645	879	753	663	389	169	63	14	28	118	312	510	746		
	CDD10.0	1127	0	0	2	13	114	216	314	277	154	36	2	0		
	CDD18.3	170	0	0	0	0	8	28	71	46	17	1	0	0		
	CDH23.3	1352	0	0	0	10	109	242	554	309	125	2	0	0		
	CDH26.7	258	0	0	0	1	22	49	124	45	17	0	0	0		
(14) Wind	WSAvg	3.7	4.2	4.2	4.1	4.4	3.8	3.4	3.0	2.8	2.9	3.5	3.9	3.8		
(15) Precipitation	PrecAvg	1082	82	59	68	82	88	103	115	110	97	103	89	89		
	PrecMax	1326	120	95	101	147	173	165	193	170	186	219	144	174		
	PrecMin	843	37	36	28	7	49	42	61	55	52	38	34	36		
	PrecStd	129	26	16	19	39	32	35	35	30	34	45	24	36		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	6.8	17.3	24.6	29.1	30.4	31.6	29.6	28.7	23.0	17.0	10.1		
		MCWB	7.1	5.0	14.4	17.1	19.4	21.3	24.2	22.5	23.0	16.5	12.8	8.3		
	2%	DB	4.5	4.4	10.4	19.6	26.4	28.1	29.6	27.9	26.5	20.4	13.5	7.2		
		MCWB	3.4	2.8	6.6	13.4	18.4	20.1	22.8	21.5	20.7	16.1	10.5	6.4		
	5%	DB	2.6	2.7	7.2	15.9	23.8	26.2	28.0	26.5	24.6	18.0	11.1	4.9		
		MCWB	1.5	1.3	4.6	10.7	16.3	19.0	21.6	20.7	19.4	14.2	9.0	4.0		
	10%	DB	0.6	1.5	5.2	13.1	21.5	24.2	26.4	25.0	22.7	15.6	9.0	2.8		
		MCWB	-0.2	0.5	3.1	8.8	15.4	18.1	20.8	19.9	18.6	12.6	7.1	1.8		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	5.7	14.4	17.4	21.8	23.4	25.4	23.8	23.5	19.3	13.7	8.6		
		MCDB	7.8	6.5	17.1	23.6	27.3	28.0	30.3	28.2	27.5	21.0	15.4	9.8		
	2%	WB	3.7	2.8	7.1	14.5	19.7	21.8	23.9	22.5	21.8	16.9	11.1	6.4		
		MCDB	4.1	3.9	9.5	17.9	24.1	26.2	27.8	26.4	24.9	19.8	12.8	7.1		
	5%	WB	1.4	1.5	5.0	11.4	18.2	20.3	22.7	21.6	20.5	14.7	9.0	4.1		
		MCDB	2.4	2.5	6.7	14.7	21.7	24.0	26.3	25.0	23.4	17.2	10.6	4.9		
	10%	WB	-0.1	0.5	3.1	9.3	16.5	19.1	21.8	20.7	19.2	13.1	7.1	1.9		
		MCDB	0.6	1.4	4.8	12.9	20.3	22.7	25.4	24.0	22.1	15.1	9.0	2.6		
(36) Mean Daily Temperature Range	5% DB	MDBR	9.7	10.0	9.5	9.8	11.7	10.9	10.9	11.0	11.6	9.5	8.0	8.0		
		MCDBR	9.3	9.5	10.9	15.5	15.5	13.4	12.1	12.3	12.8	12.8	11.3	8.3		
	5% WB	MCWBR	9.1	8.7	8.3	10.1	8.3	6.8	6.4	6.8	7.5	8.4	8.7	8.0		
		MCWBR	9.0	8.9	10.0	13.8	12.3	10.7	10.4	10.6	11.1	10.6	10.7	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.288	0.322	0.354	0.379	0.419	0.423	0.407	0.373	0.330	0.307	0.284		
	taud		2.363	2.254	2.304	2.369	2.357	2.244	2.236	2.283	2.372	2.495	2.477	2.348		
	Ebn at Noon		862	888	906	905	888	850	841	840	841	837	792	775		
	Edh at Noon		72	99	110	114	120	135	134	123	103	79	67	67		
(44) All-Sky Solar Radiation	RadAvg		1.44	2.37	3.62	4.41	5.12	5.62	5.62	4.95	3.91	2.26	1.36	0.99		
	RadStd		0.17	0.25	0.31	0.49	0.50	0.42	0.45	0.23	0.32	0.19	0.17	0.08		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page